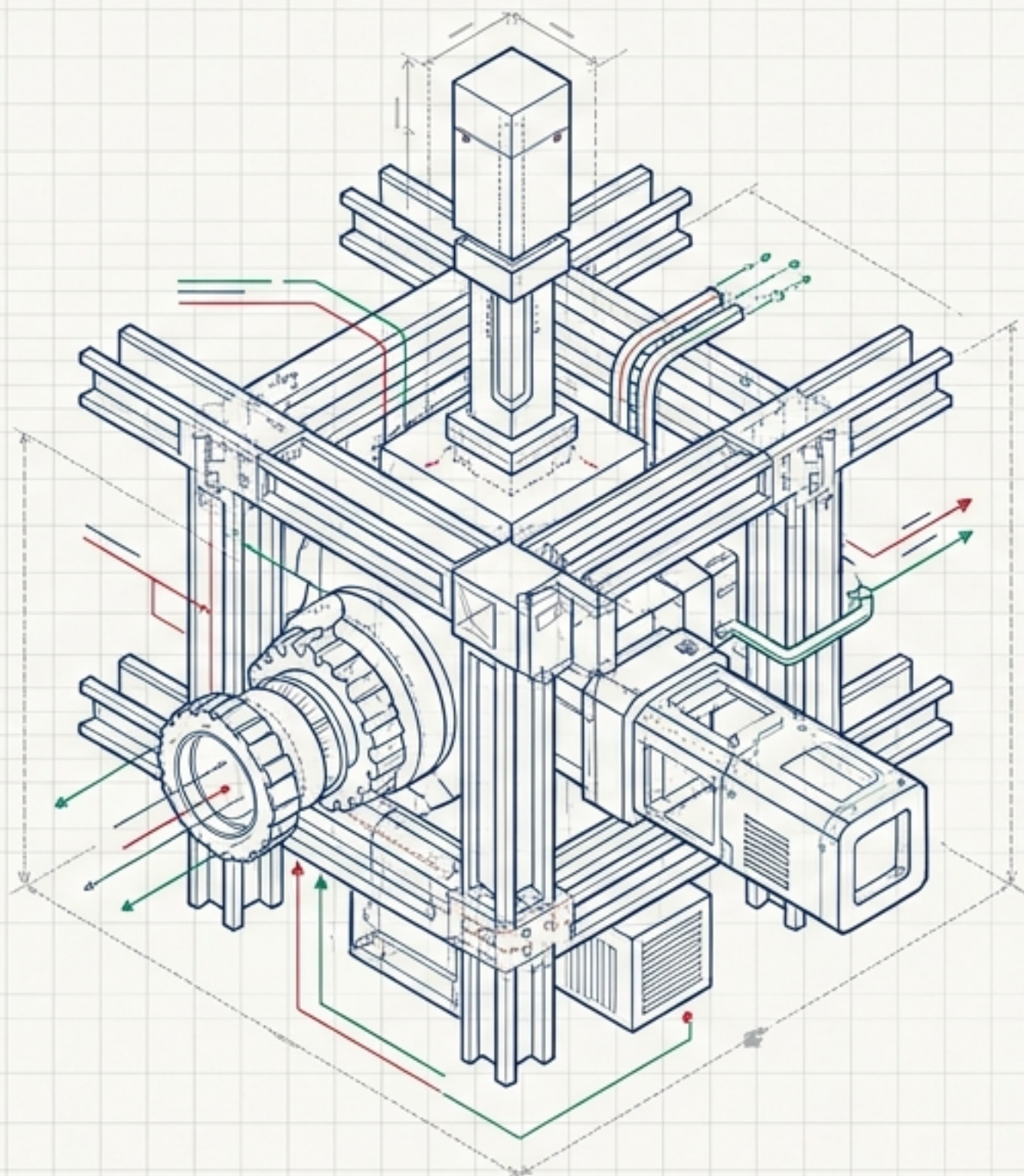


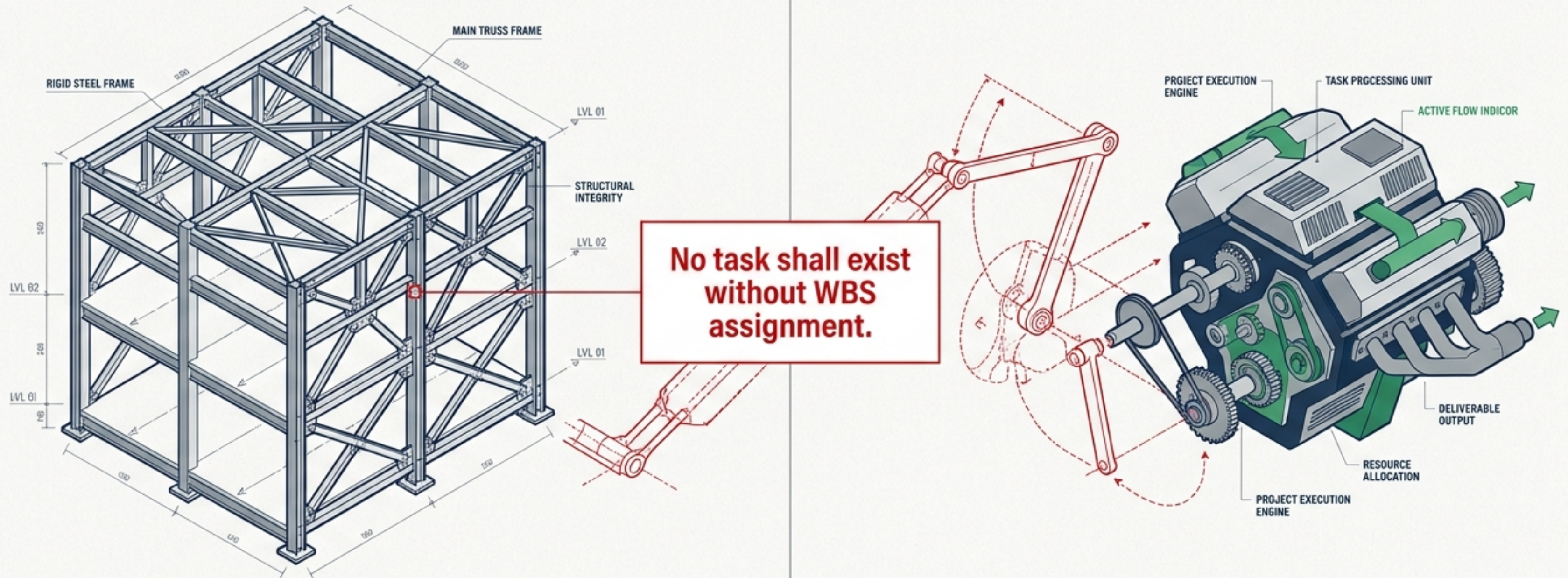


The Execution Engine

Task Management inside the Digital Construction Operating System (DCOS)

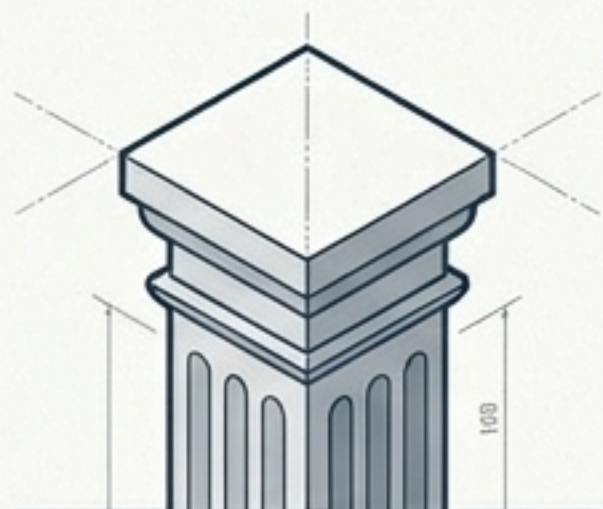
DCOS-SOP-TSK-001 | Version: 1.0

WBS is the backbone. Task Management is the heartbeat.



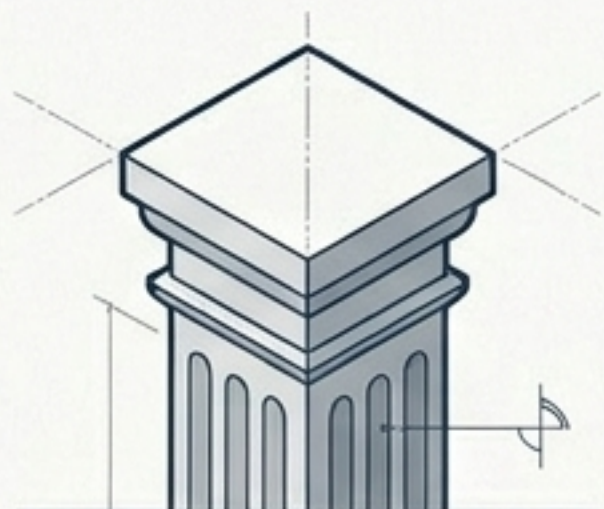
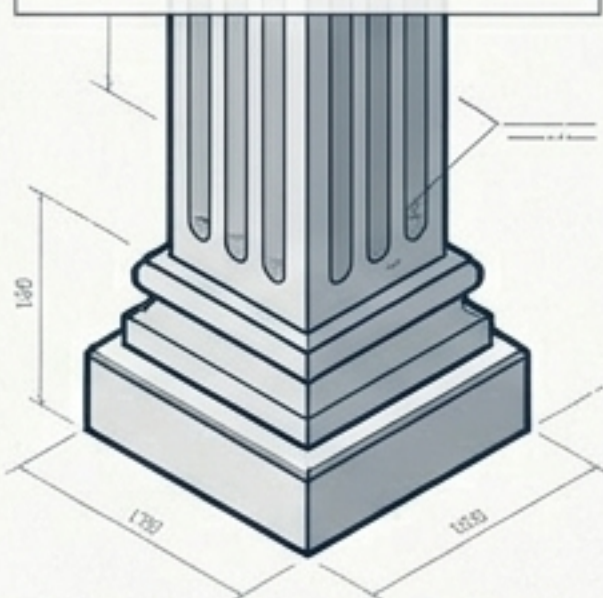
Projects are planned through WBS. Work is executed through tasks. Every deliverable, activity, review, and construction operation flows through this central process.

The 5 Pillars of DCOS Task Management



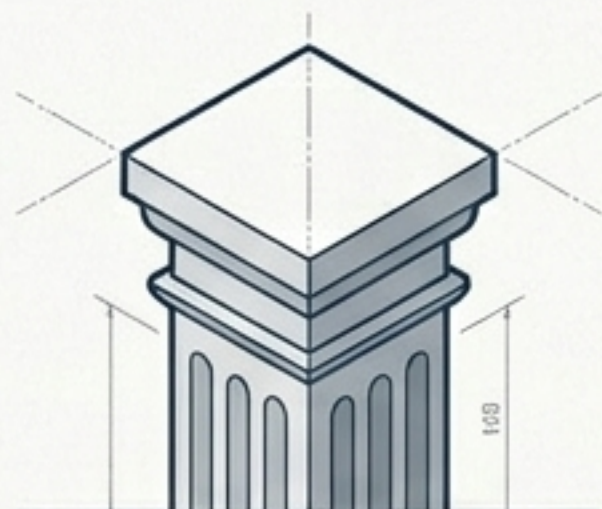
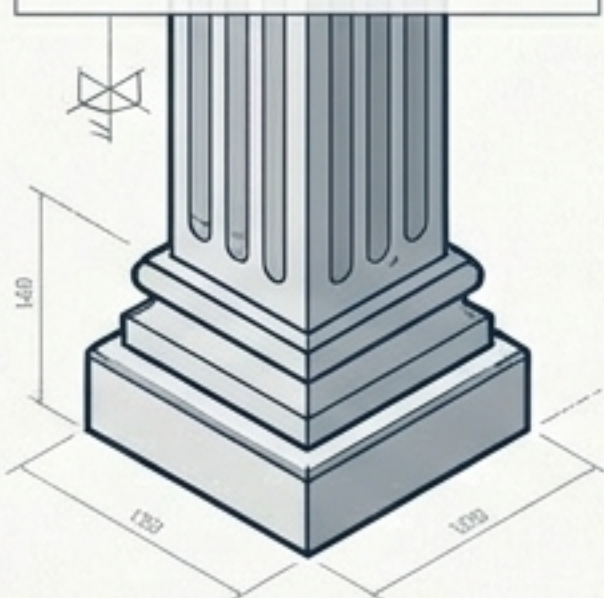
Accountability

Every task has an owner.



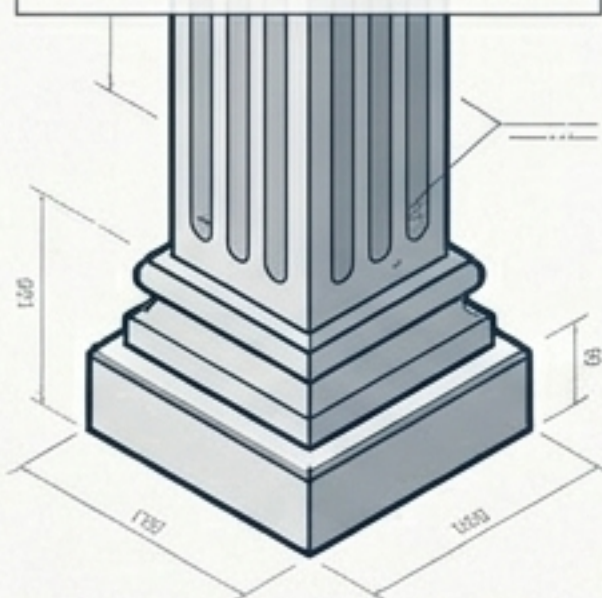
Visibility

Progress is visible in real-time.



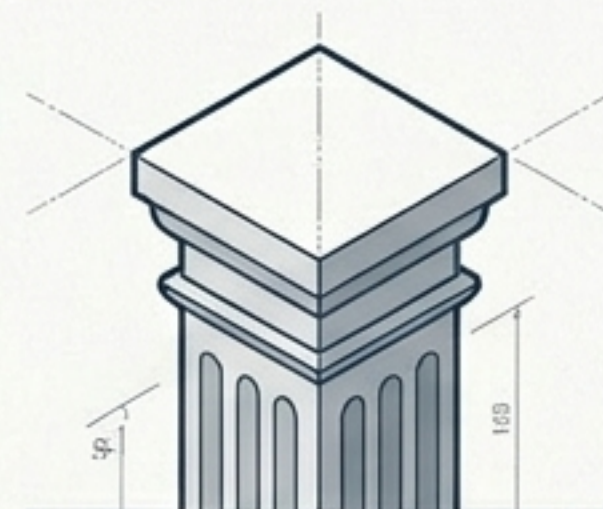
Traceability

History is permanently recorded.



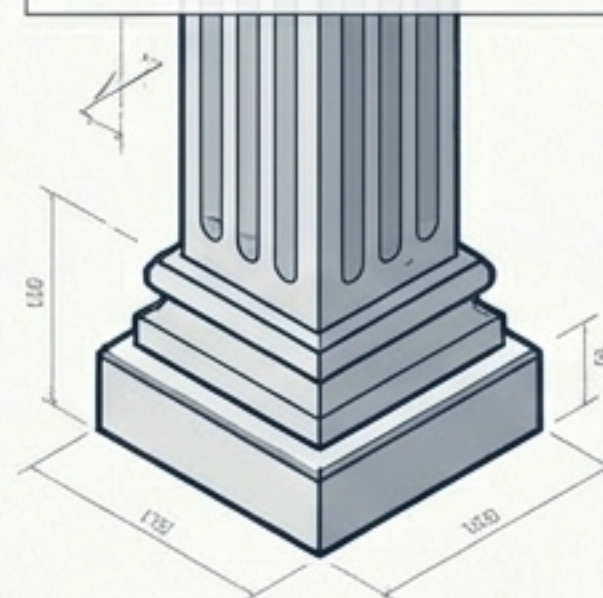
Governance

Approvals are strictly controlled.



Performance

Team productivity is measurable.

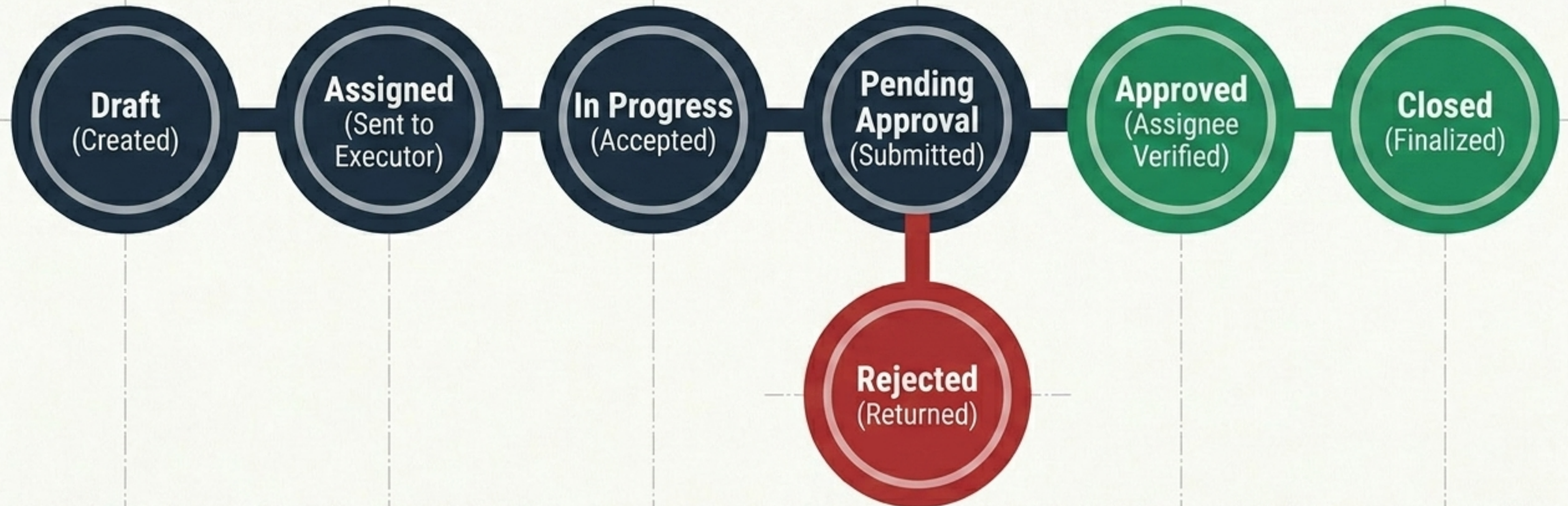


The Core Actors: Controller vs. Executor

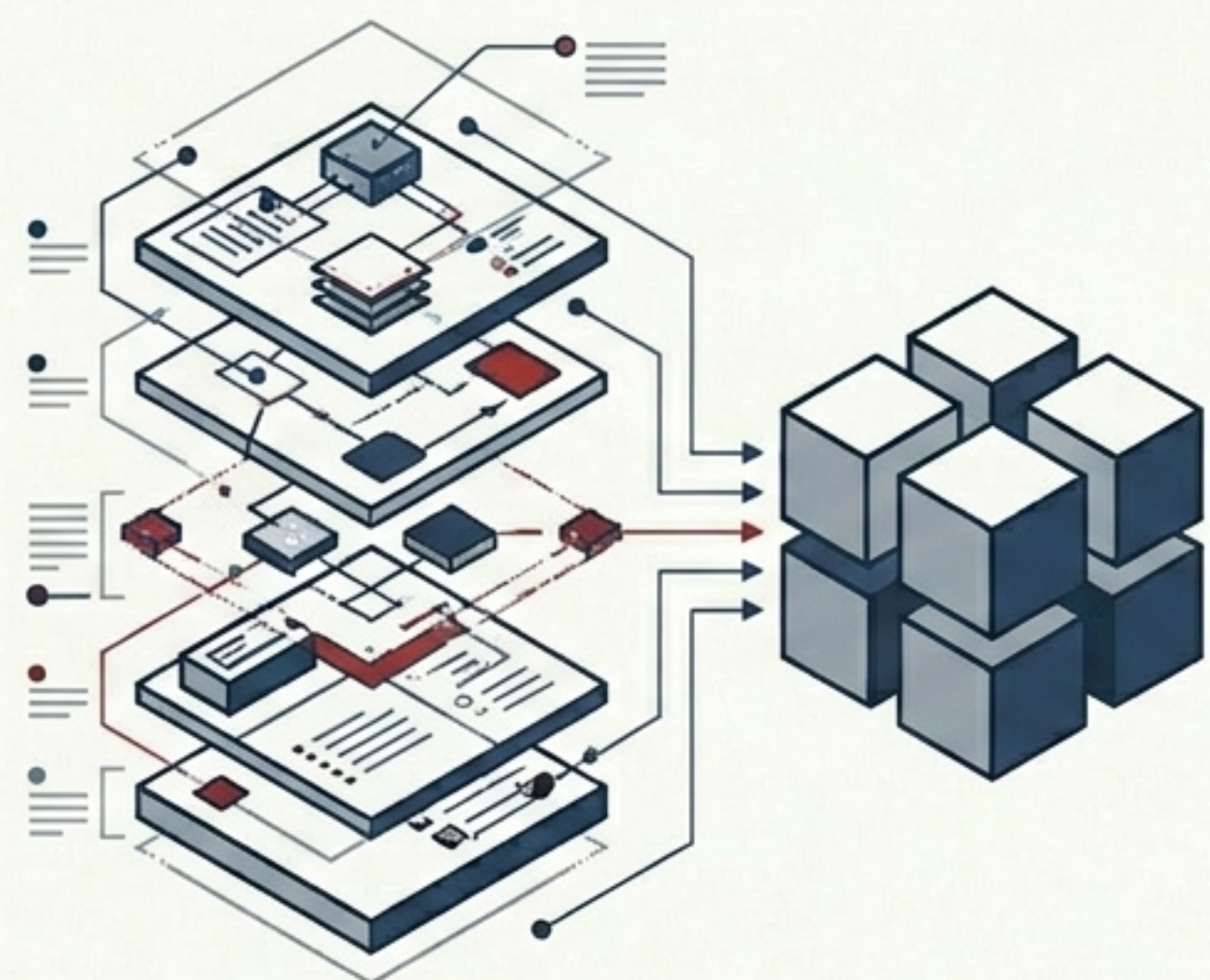
 Assignee	 Receiver
Alias: The Task Owner	Alias: The Task Executor
Role: Creates the task, sets parameters, and holds final approval authority.	Role: Accepts the assignment, performs the work, and logs progress.
Key Action: Reviews and Approves/Rejects submitted work.	Key Action: Submits completed work for final approval.

The Task Lifecycle

Applies across all task categories: Design, Procurement, Construction, QAQC, and HSE.



Generating the Node: Creation & Assignment



Every task requires WBS linkage, supporting information/attachments, and an initial Draft status. System automatically generates Recurring Tasks.

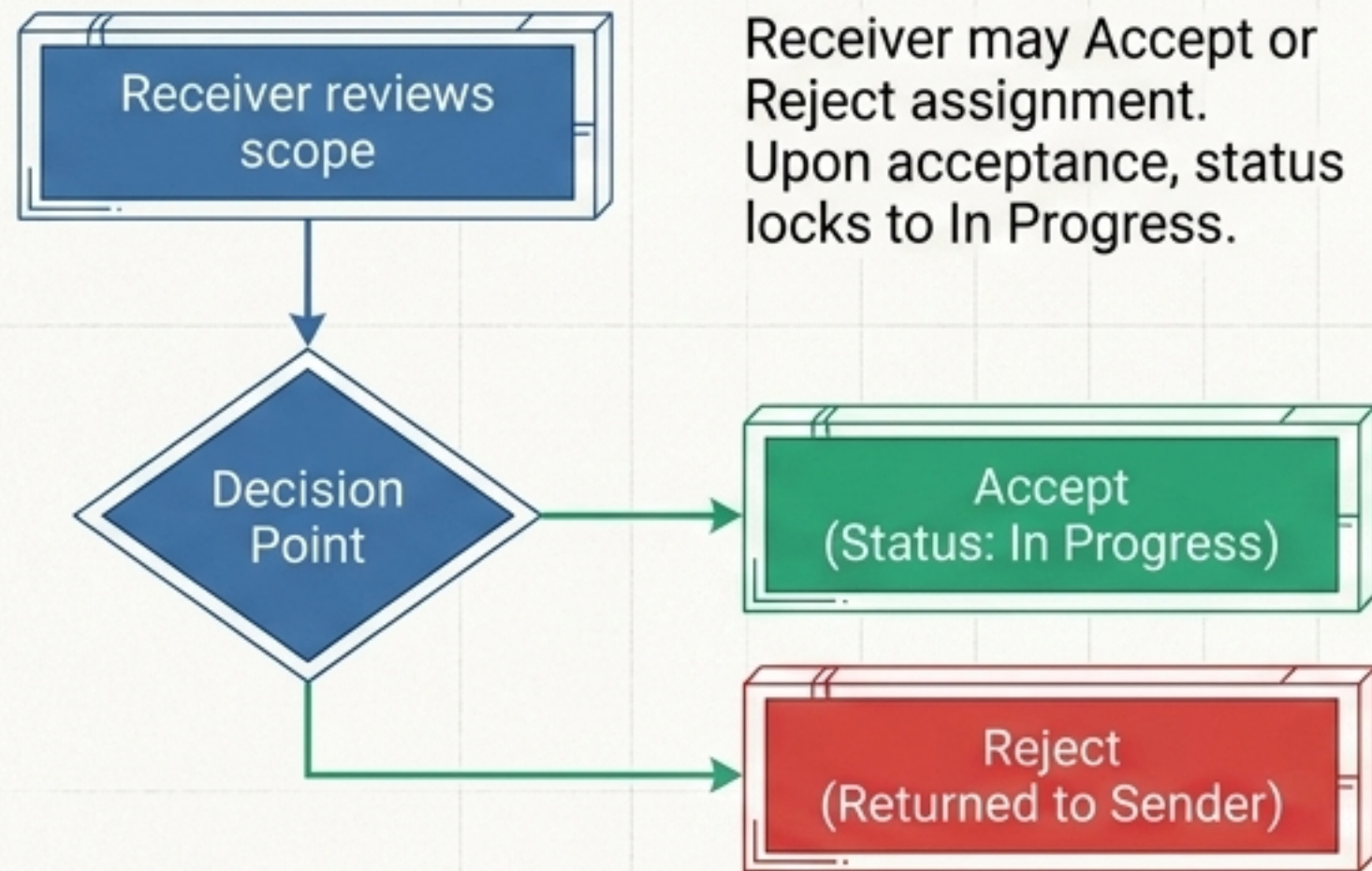
Critical	Immediate attention
High	High urgency
Medium	Standard
Low	Low urgency



The Assignment Rule: The Receiver must be explicitly defined to transition from Draft to Assigned.

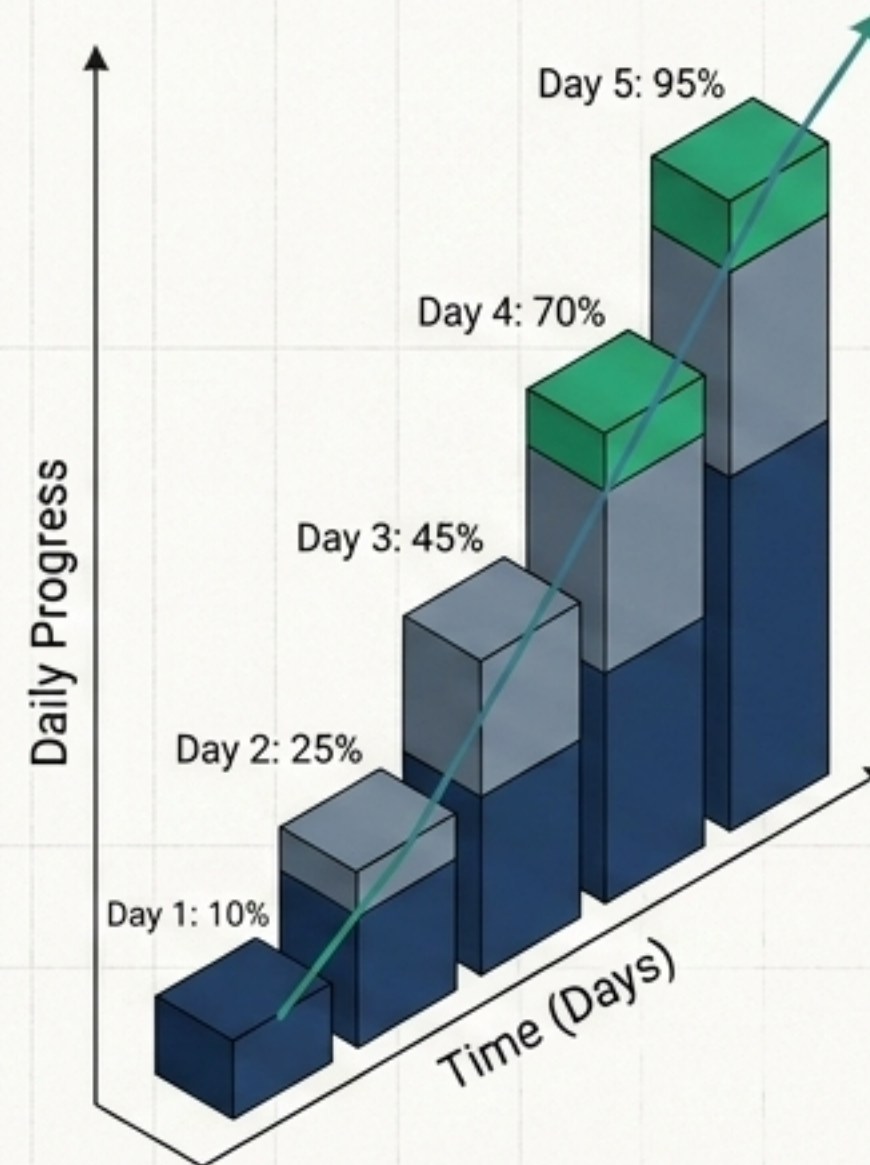
Execution & The Daily Pulse

Acceptance Protocol



Execution Rules: No task may be deleted after execution begins.

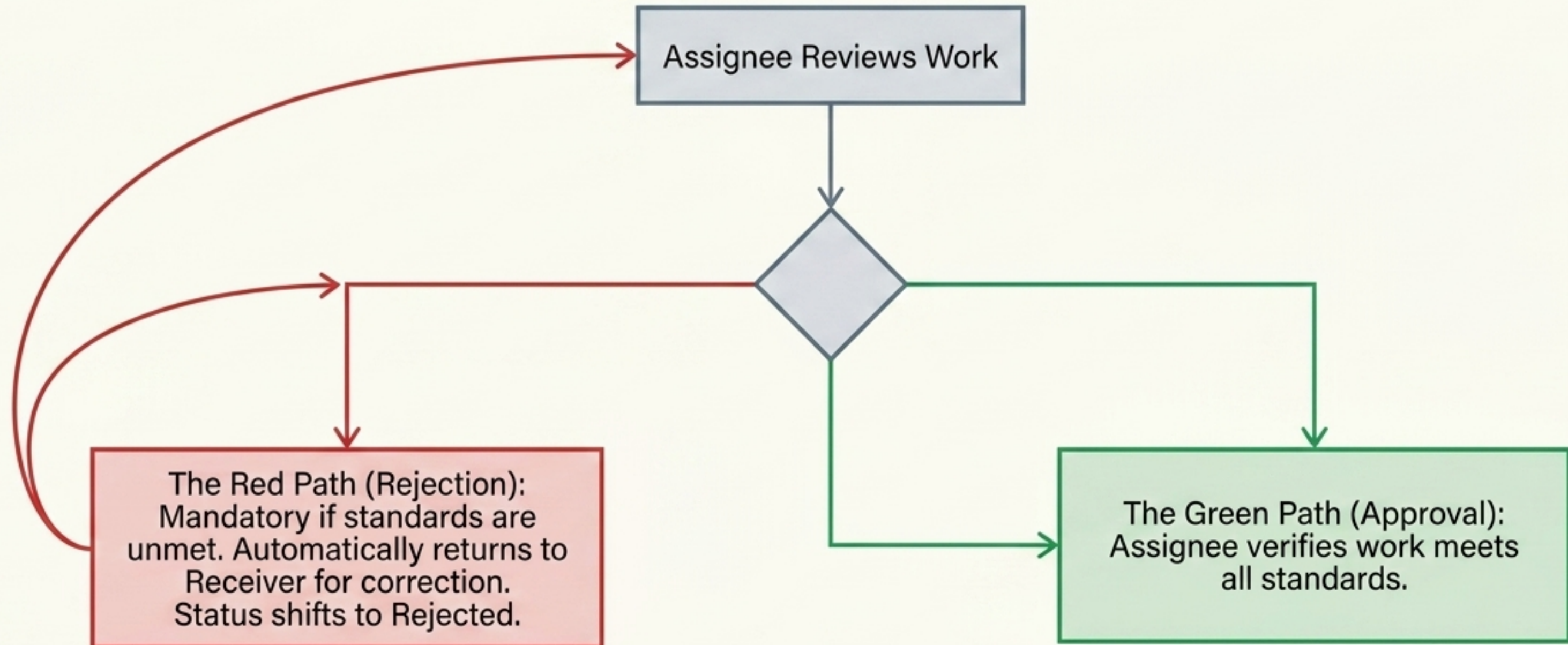
Daily Progress Update



System mandates daily updates (using time-based progress or actual percentages) for long-duration tasks to ensure real-time visibility.

The Governance Loop: Approval vs. Rejection

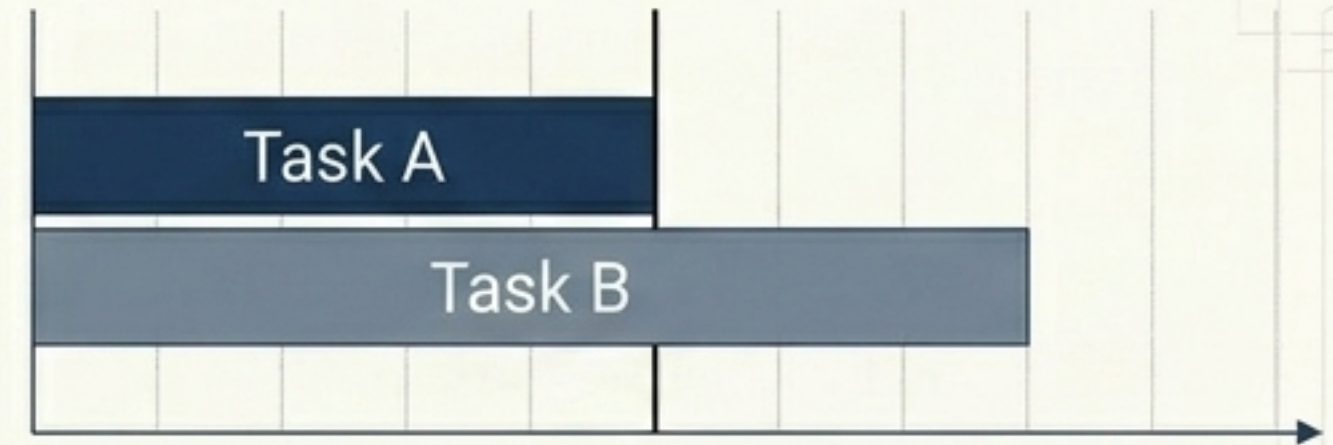
The Trigger: Work completed -> Receiver submits -> Status shifts to Pending Approval.



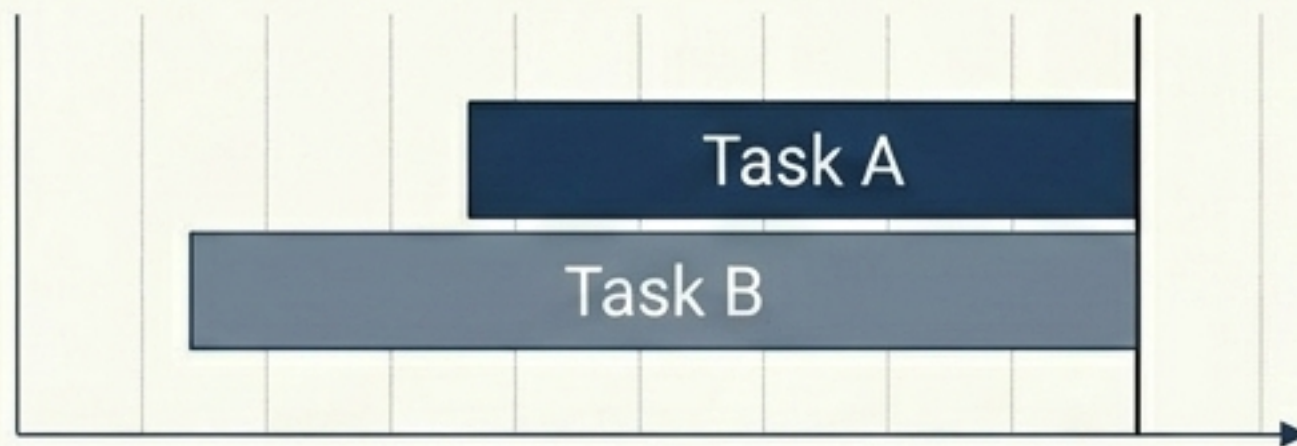
System Guardrails: Task Dependencies



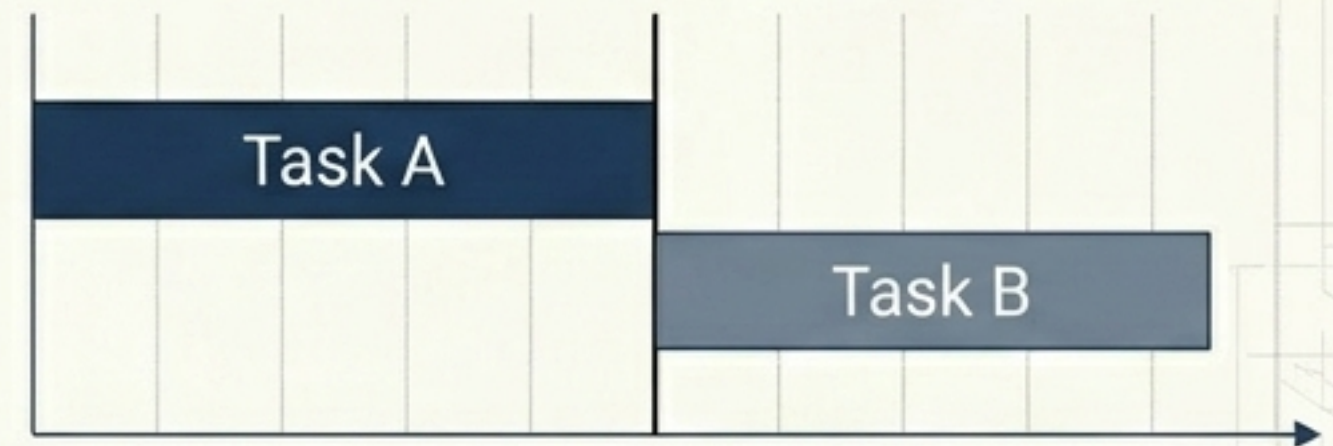
FS (Finish to Start): Task A must finish before Task B can start.



SS (Start to Start): Tasks must begin simultaneously.

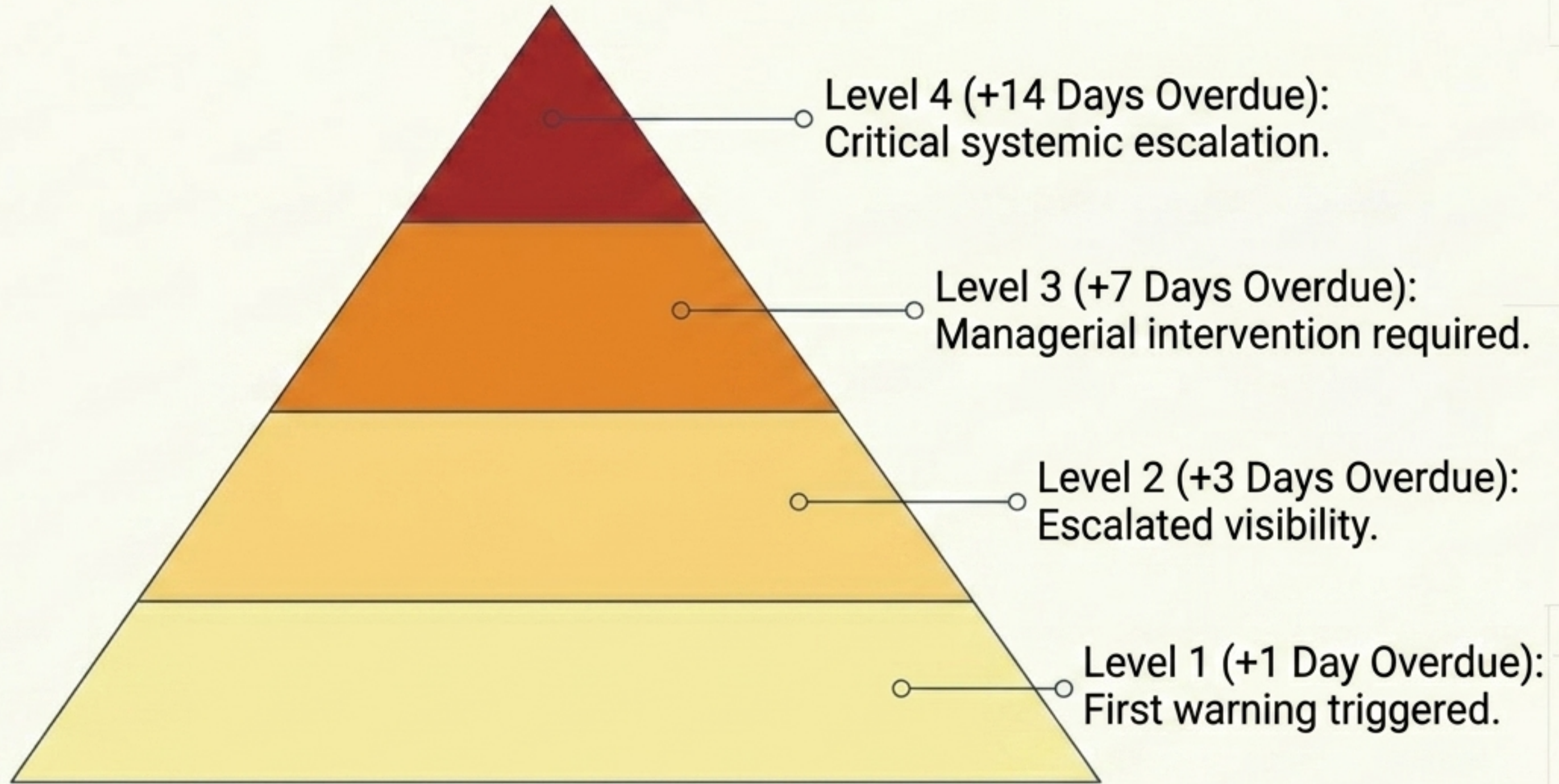


FF (Finish to Finish): Tasks must conclude simultaneously.



SF (Start to Finish): Task A must start before Task B can finish.

The Escalation Heatmap

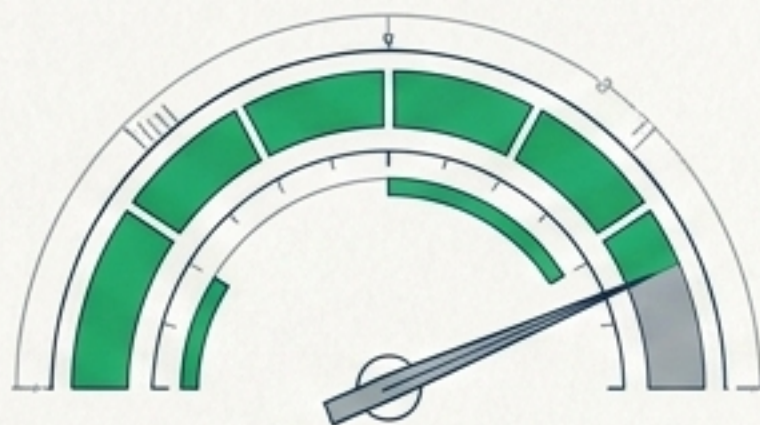


Automated Pings: The System Nervous System



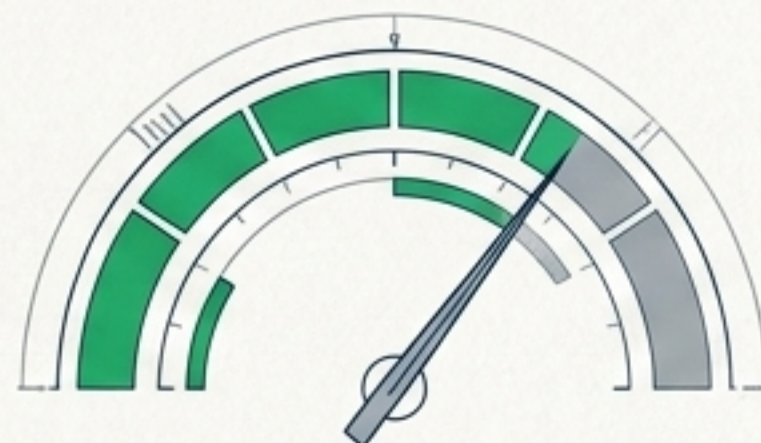
Takeaway: DCOS pushes the right data to the right actor exactly when action is required.

Measuring the Engine: System KPIs



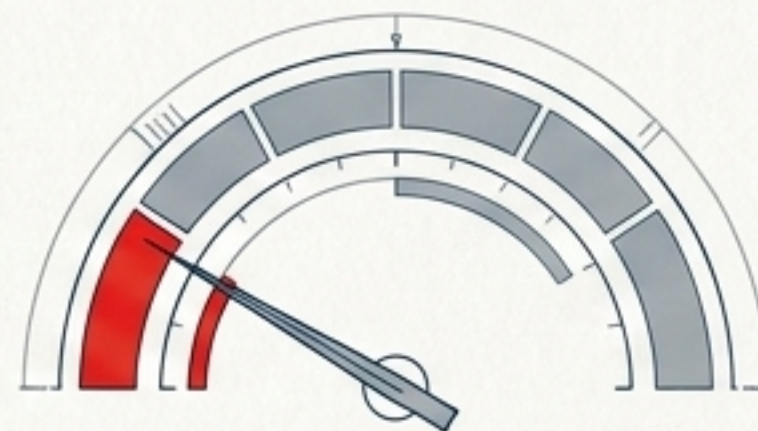
Completion Rate

Volume of work finalized vs. planned.



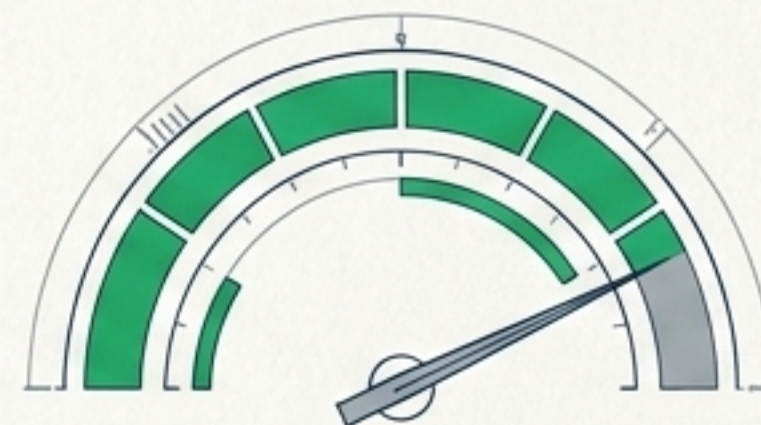
Approval Turnaround

Speed of the Assignee's governance loop.



Overdue Percentage

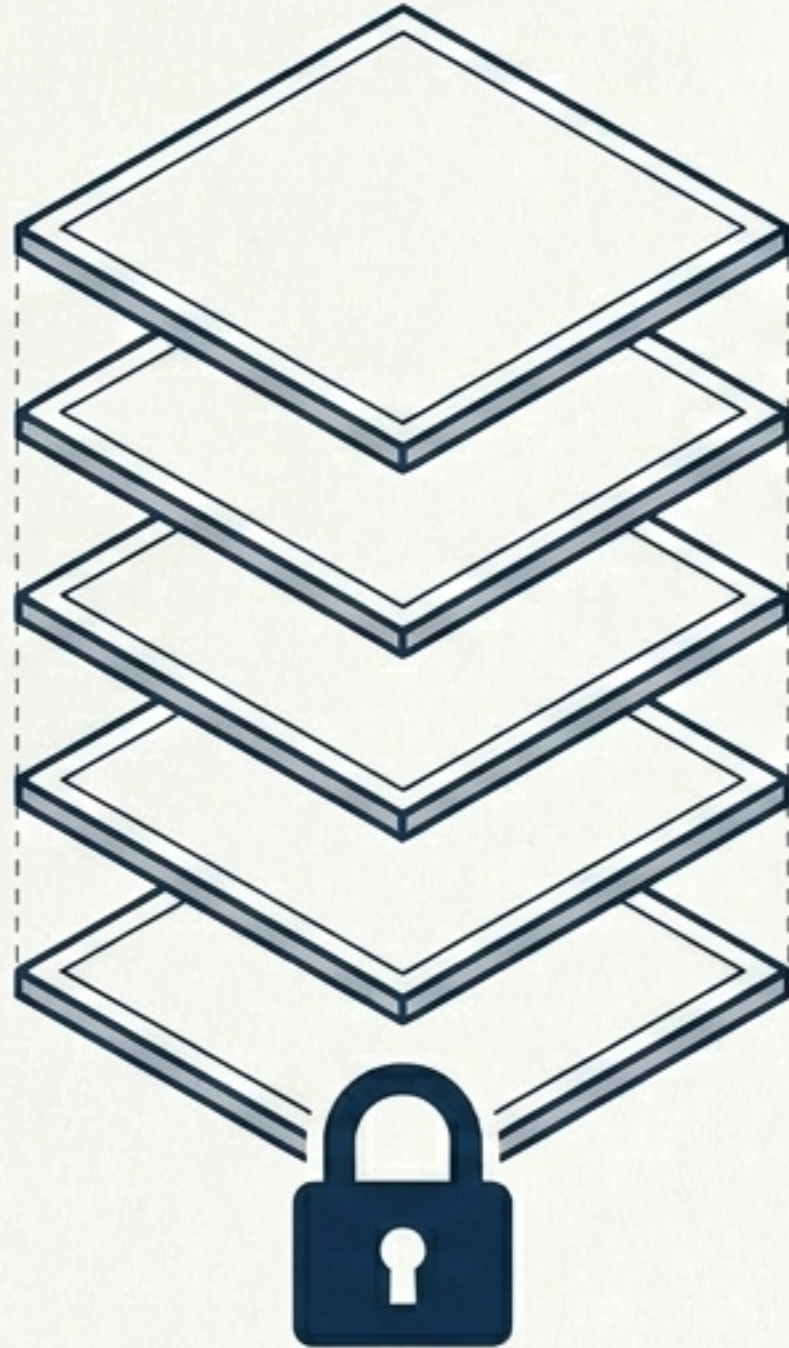
Systemic schedule health.



First-Time Approval Rate

The ultimate metric for Receiver quality and clarity of scope.

The Immutable Ledger



tasks

task_assignments

task_progress

task_comments

task_attachments

The Database Structure:

Five interconnected data layers capture every variable, comment, and file.

Audit Requirements:

Every status change, assignment, and progress update is logged with a timestamp and user ID.

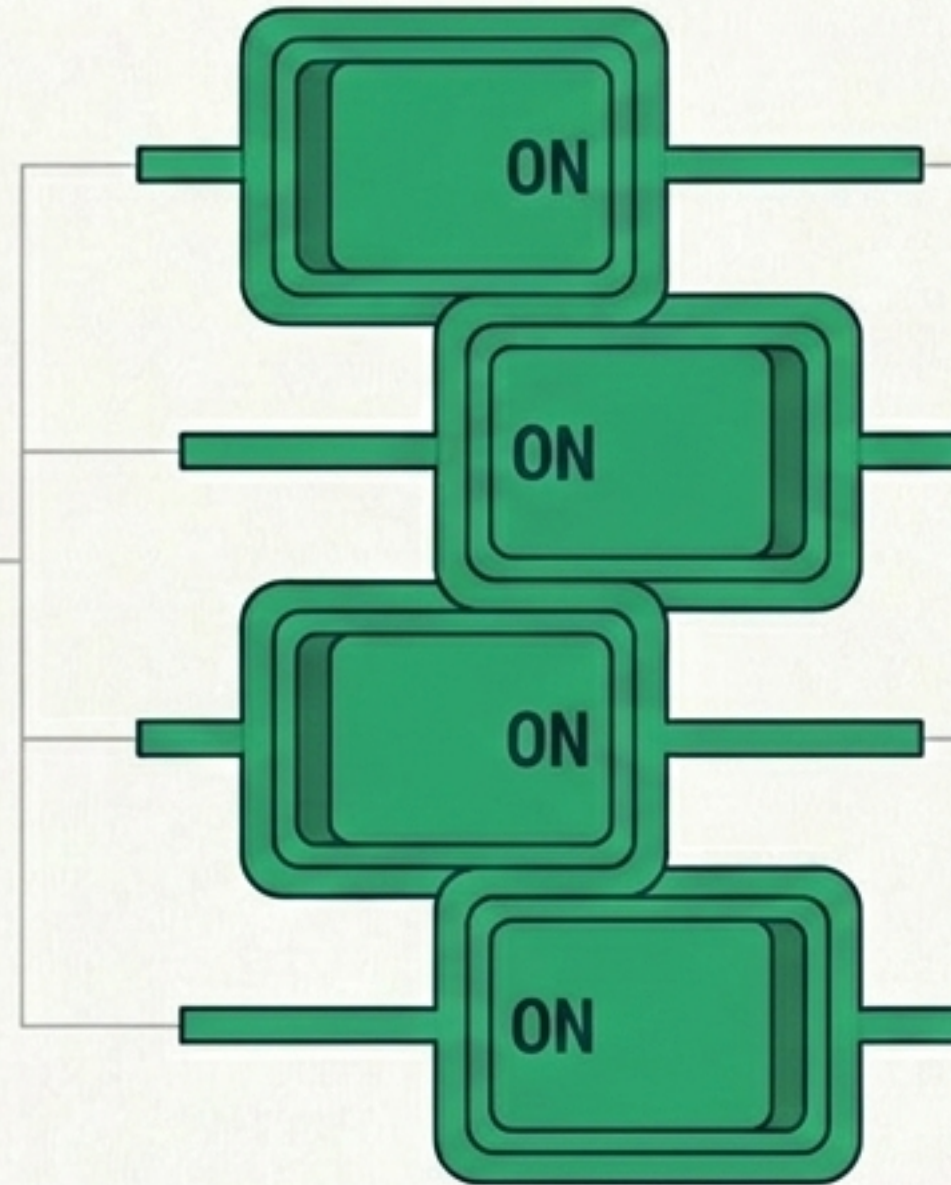
Core Security Rule:

Audit records cannot be deleted.

Closing the Loop

Context: A task is not finalized until it meets all four terminal conditions.

Condition 1:
Evaluation task is
user assigned four
own work issues.



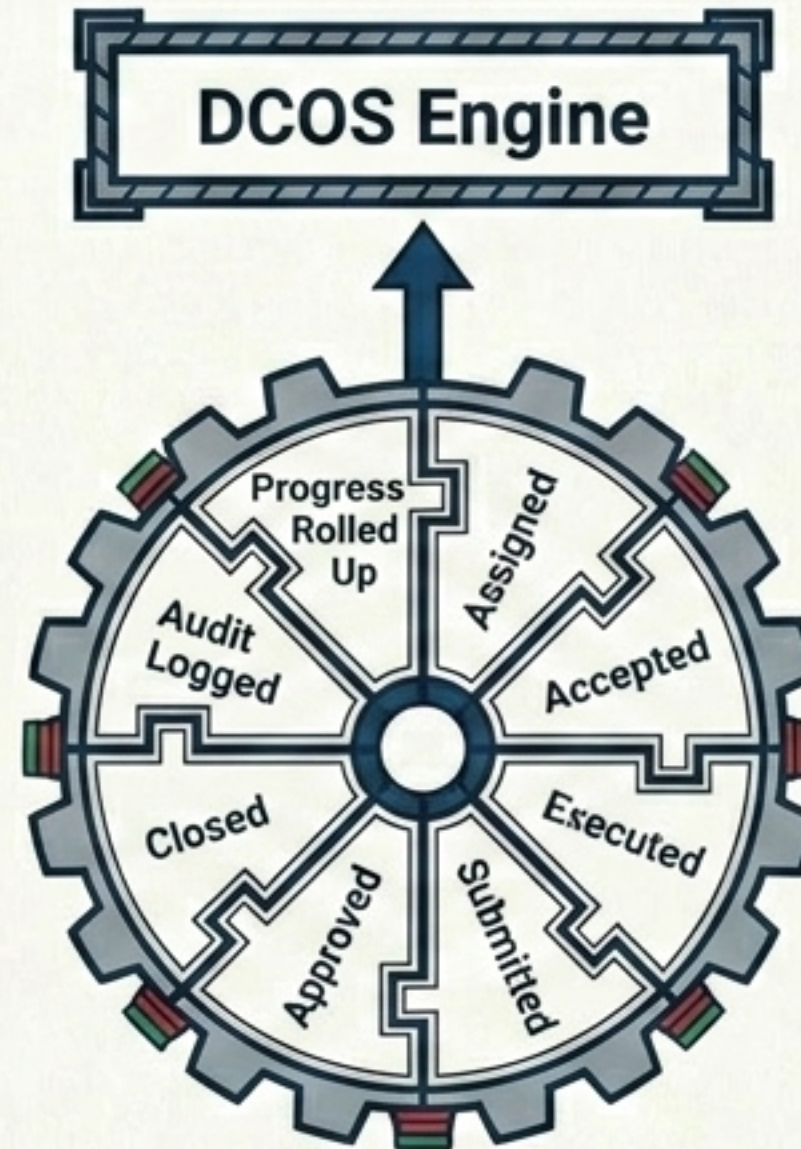
Condition 1:
Formally Approved by Assignee. ✓

Condition 2:
Progress registered at 100%. ✓

Condition 3:
All deliverables are
complete and attached. ✓

Condition 4:
Zero open issues remaining. ✓

The Definition of Done



Only when all conditions are met does a task trigger progress in the larger DCOS engine. Performance is measured through tasks. Progress is calculated through tasks. The execution engine is complete.